

Joaquin Gonzalez-Salgado

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EDUCATION

Harvey Mudd College

B.S. in Engineering (Electrical & Computer Engineering) — GPA: 3.7/4.0

Claremont, CA

Expected May 2028

EXPERIENCE

Electrical Engineer, Sponsored Engineering Project

Aug. 2026 – Present

Blue Origin

Claremont, CA

- Conducting systems trade studies of 4 sensing technologies (LiDAR, radar, Raspberry Pi computer vision, IMU) against operational requirements: cost, availability, harsh launch-environment reliability, and EMI risk to avionics
- Integrating multi-sensor position data into a closed-loop alignment controller that autonomously mates ground- and vehicle-side launch umbilical panels carrying propellant, pneumatic, hydraulic, data, and power connections
- Own component selection, sensor installation and packaging design, and hardware procurement for the full-scale model
- Work cross-functionally with Blue Origin engineers on prototype build-up, testing, and design verification

Hardware & Firmware Researcher

Aug. 2025 – Present

Analog Circuitry Engineering Lab — PI: Prof. Matthew Spencer

Claremont, CA

- Designed low-noise analog front ends with impedance matching, instrumentation amplifiers, and up to 7,000x gain, moving a dual-battery instrument to a single-supply design with integrated battery management
- Brought up and verified 4 PCB revisions in a 10-week sprint (2 single-supply analog boards, 2 nRF5340 microcontroller boards), debugging power faults with oscilloscopes, function generators, and bench supplies

PCB Designer, Electrical Team

Sep. 2025 – Present

MuddSub Autonomous Underwater Vehicle Team (RoboSub)

Claremont, CA

- Designing 2-layer power distribution and main boards in KiCad for an 8-thruster autonomous submarine, distributing 12V thruster power and regulated 5V/3.3V rails to Teensy microcontrollers and sensors
- Designing a hydrophone amplification front end using impedance matching and AC coupling to condition low-level acoustic signals for underwater localization

Electrical Engineer, Electrical & Engine Team

Jan. 2026 – Present

Harvey Mudd Racing (Formula SAE)

Claremont, CA

- Designing a modular low-voltage wiring harness of 6 subharnesses (engine sensors, injectors, ignition, cockpit, rear, shutdown circuit) around a MoTeC M130 ECU and PDM30 within a \$5.5k budget and FSAE rules
- Setting up the CAN bus between the ECU, power distribution module, and lambda sensor, and configuring ECU firmware and tuning to start the engine under electrical power

PROJECTS

NFC Business Card PCB | *KiCad, NXP NT3H2111, VNA*

Jul. 2026 – Aug. 2026

- Designed a 2-layer NFC business card around an NXP NT3H2111 tag IC with a custom PCB-trace loop antenna, impedance-matching the antenna to the 13.56 MHz NFC carrier with VNA measurements
- Laid out the card in KiCad with an LED indicator, test points, and a silkscreen QR code, and released fab-ready Gerbers, BOM, and pick-and-place files

Autonomous Underwater Vehicle (SONAR Mapping) | *Teensy 4.0, Op-Amps*

Jan. 2026 – May 2026

- Designed a 4-stage, 80 dB SONAR receive chain with a Sallen-Key bandpass filter ($f_0 = 3$ kHz) and max detector, distributing gain to preserve 121.8 kHz bandwidth, which mapped harbor depths of 1.65–2.13 m (± 0.15 m)

36V Electric Vehicle | *Power Distribution, Circuit Protection, Raspberry Pi*

Jan. 2026 – May 2026

- Led a team building a 36V electric vehicle that holds a steady 15 mph by designing its power distribution and protection system (motor controllers, relays, fuses, breakers)

TECHNICAL SKILLS

Hardware: Analog Front Ends, PCB Layout & Bring-Up, Power Distribution, BMS, nRF5340, STM32, ESP32

Languages & Tools: Python, C, C++, MATLAB, Verilog, SystemVerilog, VHDL, Git, CMake, Jupyter

Engineering Software: Altium, KiCad, LTspice, Lattice Radiant, Questa, Quartus, LabVIEW, Fusion CAD

Firmware & Embedded: Zephyr RTOS, Bare-Metal C, I2C, SPI, UART, CAN, Bluetooth, ADC, PWM, IoT

Controls: Feedback Control, PID Control Loops, Motor Speed Control, Sensor Integration & Calibration

Lab Instrumentation: Oscilloscope, Multimeter, Function Generator, Power Supply, VNA, Data Acquisition